

BSc graduation project, University of Jeddah · phase B, the simulator against the car

The simulator against the Supra's logs

The engine simulator set beside the logged drives of the team's test car, a Toyota GR Supra. Five comparisons, one chart each, ordered from the strongest kind of evidence to the weakest.

82,984 logged samples · 7 drives · 26 steady points · gearbox ZF 8HP51 · engine B58, 3.0 L

INDEPENDENT CHECK	The gearbox: the ratios the car ran, against the eight Toyota publishes 86.7 % of moving samples within ±4% of a published ratio
INDEPENDENT CHECK	Boost pressure under boost: the model against the car's own boost channel medians +1.9 % apart; with the old sensor reading, +25 %
FIT	Boost against air flow: the simulator's ceiling on the car's envelope RMS 0.135 in pressure ratio; the top bins rest on a handful of readings each
FIT	Enrichment: the mixture under boost, against the model's map largest cell gap 0.076 in λ
CONSISTENCY CHECK	Engine load at the steady points: the model against the ECU's load channel 1.4 % with nothing fitted -- and it cannot test the breathing model
INDEPENDENT CHECK	the model side came from a published source, or from a model written without this data; the logs never touched it
FIT	the model side was fitted to this car's logs; the chart shows the fit and tests nothing
CONSISTENCY CHECK	the comparison cancels the model it appears to test (CLAUDE.md mistake 12)

Not charted, and why

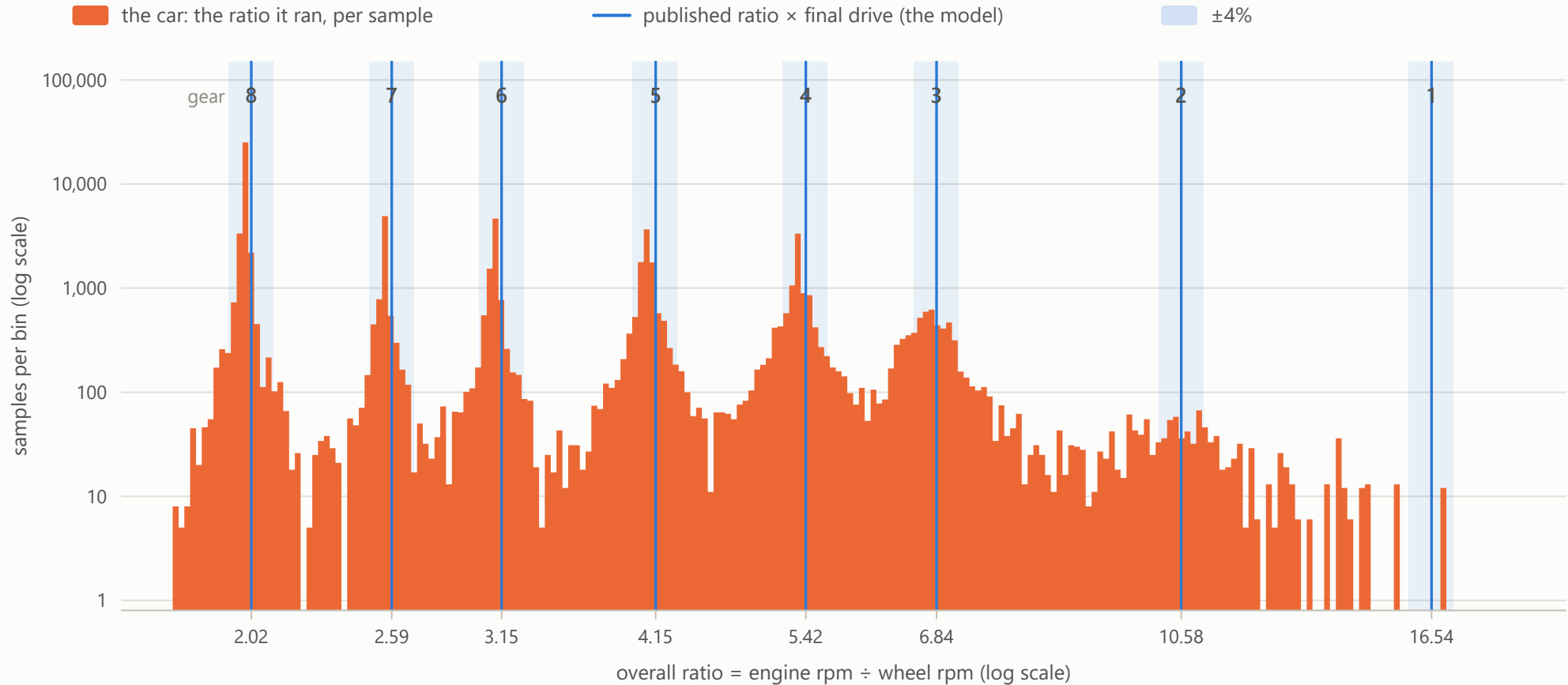
- Turbine housing temperature, the quantity the project protects: the car has no sensor for it.
- Knock: the model's knock integral and the car's own retard are unrelated (AUDIT.md H5). A negative result, and no script charts it yet.
- Oil temperature over a whole drive, modelled against measured: not computed yet.
- Air mass flow and coolant: equal by construction, so a chart of them would prove nothing.

The gearbox: the ratios the car actually ran, against the eight Toyota publishes

79,134 moving samples (above 20 km/h) from 7 drives · 86.7 % lie within $\pm 4\%$ of a published ratio

INDEPENDENT CHECK

the ratios and the final drive are Toyota's published figures; nothing was fitted to these logs



Overall ratio = engine rpm ÷ wheel rpm, the wheel's rpm taken from road speed and the model's wheel radius (0.33 m, whose source is not recorded). The radius shifts all eight peaks together, so the SPACING of the peaks is the independent part. Samples between the peaks are gear changes and torque-converter slip, which the model does not simulate: it assumes the converter locked. The car's own 'Actual gear' channel stops at 6, so it is not used (CLAUDE.md mistake 18). Data: data/master_samples.csv.

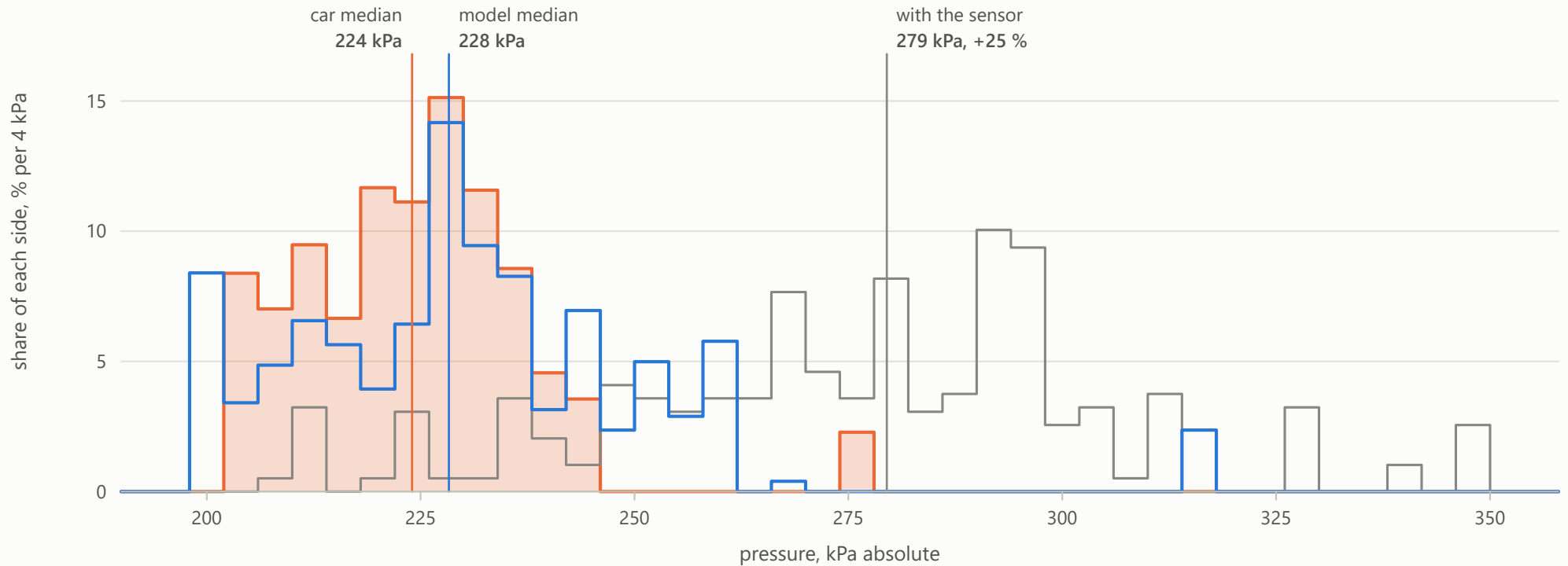
Boost pressure under boost: the model's value against the car's own boost channel

above 200 kPa · model 762 samples, car 1,097 logged rows · medians 228 and 224 kPa · gap +1.9 %

INDEPENDENT CHECK

plant.charge_temperature() was written without this data and has no parameter fitted to it

the car: its Boost pressure channel the model: inverted with charge_temperature() the old error: inverted with the sensor



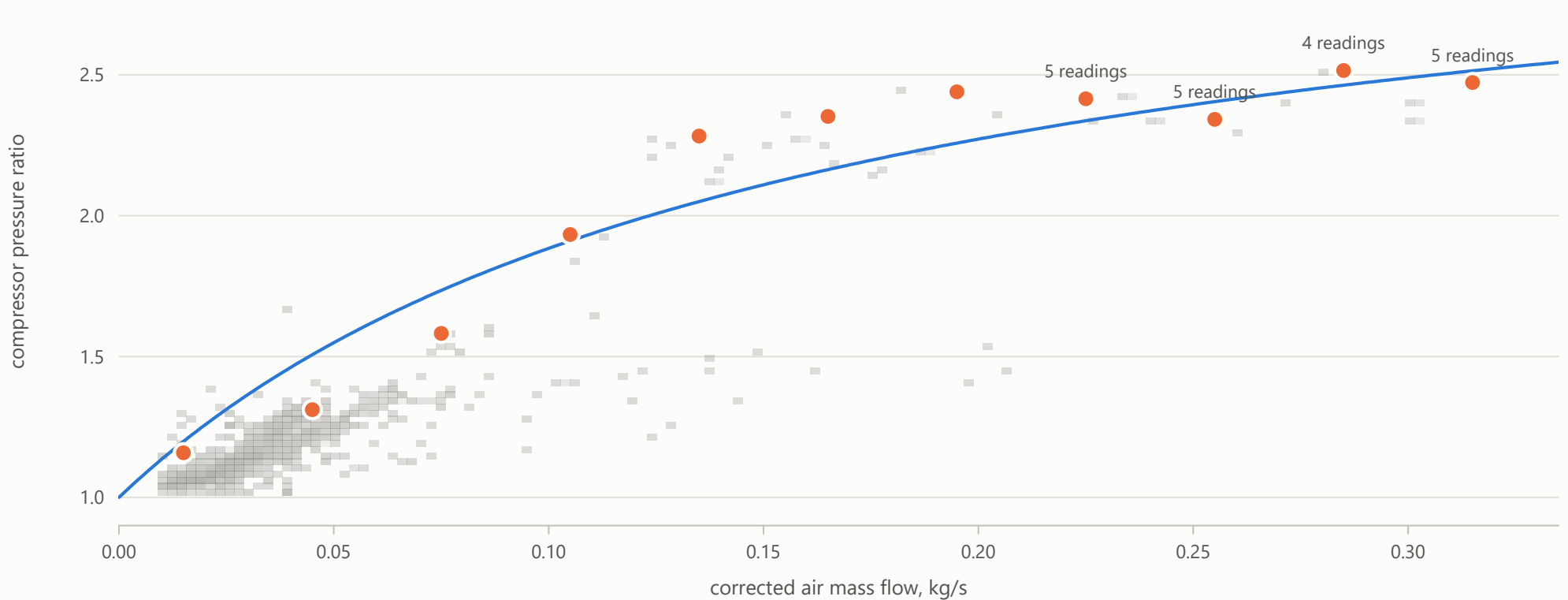
The two sides are different samples, not pairs: the logger reads one channel per row, so air mass and boost are never read at the same instant, and two populations over the same region are compared by their medians. Model: manifold pressure inverted from the measured air mass, with the modelled charge temperature. Grey: the same samples inverted with the pre-throttle sensor, which is a compressor outlet, not the charge -- the dataset's error until 10 September (CLAUDE.md mistake 13). Samples at the air-mass sensor's ceiling are excluded (mistake 7). Data: data/master_samples.csv, logs/raw/*.csv.

Boost against air flow: the simulator's ceiling on the car's measured envelope

40,716 steady samples · 95th percentile per 0.03 kg/s bin · RMS gap 0.135 in pressure ratio

FIT the ceiling (plant.boost_ceiling_kpa) was fitted to this car's envelope; the chart shows the fit and tests nothing

the car: steady samples the car: 95th percentile per bin the model: boost ceiling



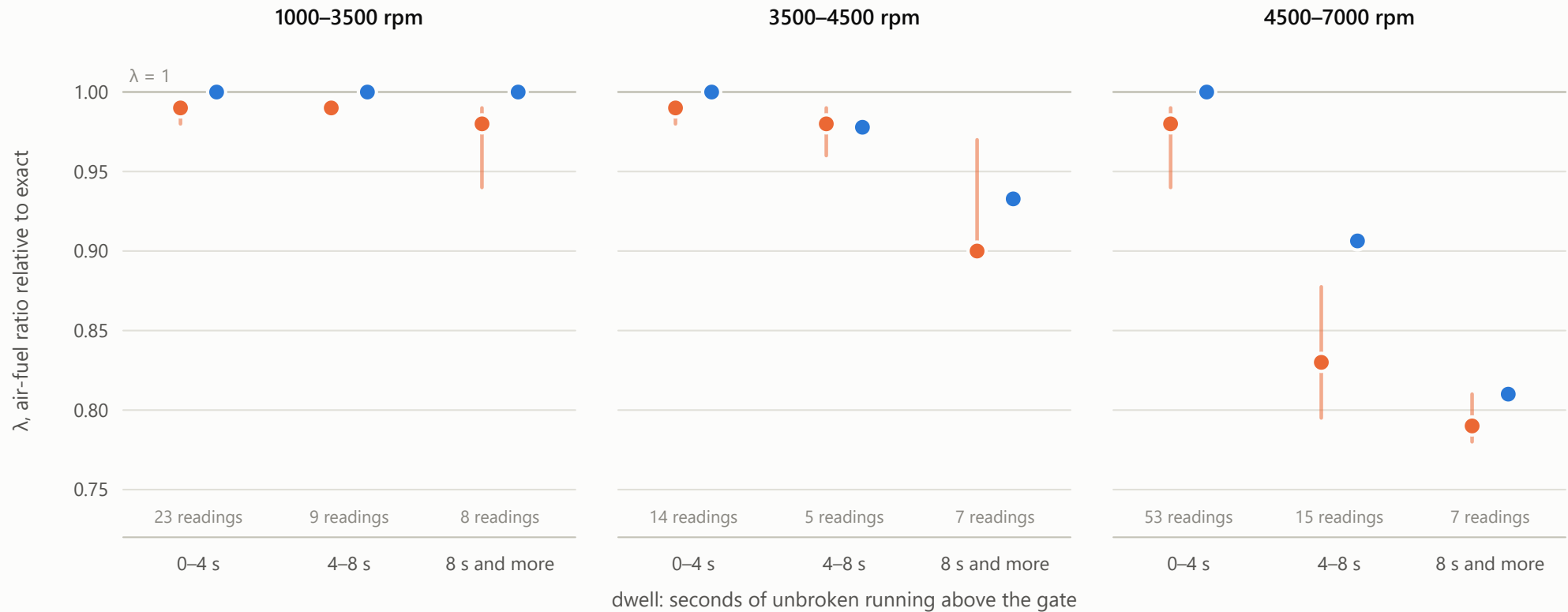
Grey: every steady sample with the air-mass sensor below its ceiling; darker means more samples. Orange: the 95th percentile of pressure ratio in each bin (fit_envelope.py). Read the counts on the top bins: the rows are forward-filled, and the number beside each dot is how many INDEPENDENT readings it rests on (AUDIT.md H4). Beyond the last bin nothing is measured, because the air-mass sensor saturates (CLAUDE.md mistake 7). Data: data/master_samples.csv.

Enrichment: the mixture the car ran under boost, against the model's map

1,341 samples above the 180 kPa gate · nine cells by engine speed and seconds of dwell · largest gap 0.076 in λ

FIT base_lambda v4 was fitted to these same samples (engine_env.BaselineECU); it tests nothing

● the car: median, and its middle half ● the model: base_lambda on the same samples



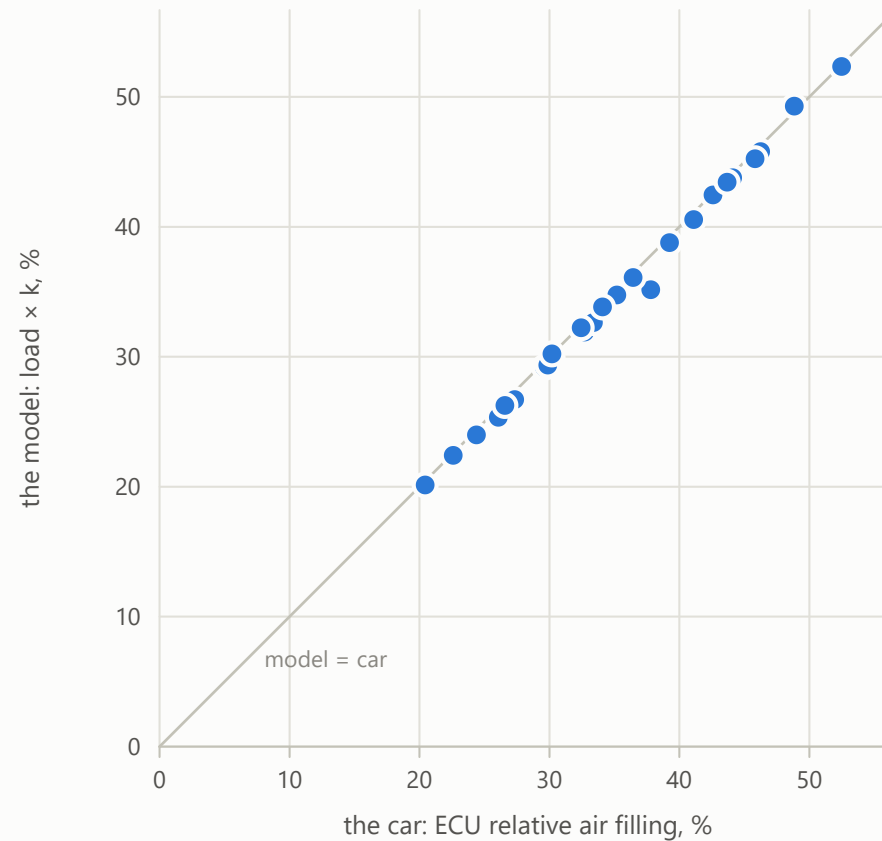
$\lambda = 1$ is the chemically exact mixture; below 1 is rich, fuel spent to cool the exhaust. Dwell = seconds of unbroken running above the gate, from the timestamps (verify_docs.dwell_column). Dots are medians over the cell; the orange bar is the middle half of the car's samples; the count under each cell is how many independent lambda readings it rests on (AUDIT.md H4). The map's dwell constants were fitted before AUDIT.md H3 corrected the dwell axis, and the worst cell shows it: 4500–7000 rpm at 4-8 s, car 0.83 against model 0.91. The locked scenario's climb runs in the first panel's band, where the map never enriches. Data: data/master_samples.csv.

Engine load at the steady points: the model against the ECU's own load channel

26 points, 30–75 kPa manifold pressure · mean error 1.4 % with nothing fitted · 1.1 % with one fitted constant

CONSISTENCY CHECK

not a test of the breathing model: η_v , residual gas and intake temperature cancel out



Horizontal: the ECU's 'Relative air filling'. Vertical: the model's load from the measured air mass, times $k = 269.6 / T_{\text{charge}}$ -- the DIN reference state, 1013 mbar and 0 °C -- with nothing fitted. Both sides are built from the same air-mass reading, so the agreement shows what the ECU channel means and that the model has the right displacement (a wrong engine misses badly); it cannot show that the model breathes like the car (CLAUDE.md mistake 12). 'Steady' means steady in road and engine speed only (AUDIT.md M3). Data: data/master_points.csv.

The numbers behind each chart

1. The gearbox: the ratios the car ran, against the eight Toyota publishes

gear	published overall ratio	samples within $\pm 4\%$
1	16.537	12
2	10.584	348
3	6.842	3,598
4	5.418	7,551
5	4.145	9,187
6	3.150	8,189
7	2.589	7,314
8	2.016	32,376

2. Boost pressure under boost: the model against the car's own boost channel

side	samples	median, kPa	gap to the car
the car: Boost pressure channel	1,097	224.0	--
the model: charge_temperature()	762	228.3	+1.9 %
the old error: the pre-throttle sensor	587	279.5	+24.8 %

3. Boost against air flow: the simulator's ceiling on the car's envelope

flow bin, kg/s	car p95 PR	model PR	rows	independent readings
0.015	1.159	1.198	24,060	1,120
0.045	1.312	1.507	14,759	809
0.075	1.582	1.735	1,040	68
0.105	1.933	1.911	166	15
0.135	2.283	2.050	249	22
0.165	2.353	2.164	142	15

0.195	2.440	2.258	124	13
0.225	2.415	2.337	36	5
0.255	2.342	2.405	40	5
0.285	2.515	2.463	53	4
0.315	2.473	2.514	47	5

4. Enrichment: the mixture under boost, against the model's map

cell	rows	independent readings	car median	car middle half	model
λ cell: 1000–3500 rpm, 0-4 s	249	23	0.99	0.98 to 0.99	1.00
λ cell: 1000–3500 rpm, 4-8 s	123	9	0.99	0.99 to 0.99	1.00
λ cell: 1000–3500 rpm, 8+ s	69	8	0.98	0.94 to 0.99	1.00
λ cell: 3500–4500 rpm, 0-4 s	137	14	0.99	0.98 to 0.99	1.00
λ cell: 3500–4500 rpm, 4-8 s	45	5	0.98	0.96 to 0.99	0.98
λ cell: 3500–4500 rpm, 8+ s	53	7	0.90	0.90 to 0.97	0.93
λ cell: 4500–7000 rpm, 0-4 s	435	53	0.98	0.94 to 0.99	1.00
λ cell: 4500–7000 rpm, 4-8 s	142	15	0.83	0.80 to 0.88	0.91
λ cell: 4500–7000 rpm, 8+ s	88	7	0.79	0.78 to 0.81	0.81

5. Engine load at the steady points: the model against the ECU's load channel

rpm	manifold kPa	ECU load, %	model load × k, %	error, %
2439	30	20.4	20.1	−1.5
2931	33	22.6	22.4	−0.8
2582	36	24.4	24.0	−1.6
1552	40	26.1	25.3	−2.8
2443	40	26.6	26.3	−1.2
1879	42	27.3	26.7	−2.3

2039	45	29.6	29.2	-1.4
3178	44	29.9	29.4	-1.7
5053	45	30.2	30.2	+0.1
2615	44	30.2	29.8	-1.4
4184	47	32.4	32.2	-0.6
1553	50	32.7	31.9	-2.5
1707	50	33.4	32.6	-2.2
2935	49	34.1	33.8	-0.7
1878	53	35.2	34.8	-1.2
2615	53	36.4	36.1	-0.9
1549	54	37.8	35.2	-7.0
2039	58	39.2	38.8	-1.2
1876	61	41.1	40.6	-1.3
2614	61	42.6	42.5	-0.3
2918	61	43.7	43.4	-0.5
2882	62	44.1	43.8	-0.7
2260	66	45.8	45.2	-1.3
2040	67	46.3	45.8	-1.0
1879	73	48.8	49.3	+0.9
2274	75	52.5	52.3	-0.3

Built by `plot_sim_vs_car.py` from `data/master_samples.csv`,
`data/master_points.csv` and `logs/raw/`. Every value on this page is printed,
with its definition, in `figures/sim_vs_car/numbers.txt`. Read-only: nothing in
this project writes to the car.